Probabilistic Recurrent Intention Switching Model

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Abstract

Inverse reinforcement learning (IRL) recovers reward functions from observed behavior, yet traditional methods assume a single stationary reward that cannot capture goal switching within an episode. Recent multi-intention IRL methods address this by segmenting trajectories, but model intention transitions as either a memoryless Markov chain or via manual state augmentation with a fixed history window. We propose the Probabilistic Recurrent Intention Switching Model (PRISM), which replaces both mechanisms with a lightweight recurrent network that maps observation history to a per-step intention distribution. We prove that the resulting EM objective decomposes exactly into independent per-intention reward subproblems, each solvable in closed form, yielding an $\mathcal{O}(nK)$ E-step with no variational approximation. We evaluate PRISM on a non-Markovian gridworld, a mouse labyrinth, and BridgeData V2 robotic manipulation, the first large-scale robotic application of multi-intention IRL. Across all settings PRISM achieves the highest held-out log-likelihood while recovering nameable, temporally coherent intentions from unlabeled demonstrations, suggesting that discrete goal switching is present in both biological and artificial agents.

1 Introduction

Animals and robots frequently pursue multiple goals within a single episode. A mouse navigating a labyrinth may alternate between seeking water and returning home; a teleoperated robot arm may switch between reaching, grasping, and placing within one demonstration. Understanding these behaviors requires recovering not only what rewards drive each goal, but also when the agent switches between them and what aspects of the history trigger these switches.

Inverse reinforcement learning (IRL) addresses the first question by inferring a reward function from observed trajectories. Standard IRL assumes a single stationary reward, conflating multiple objectives into one function that cannot distinguish between them. Three lines of work have attempted to lift this stationarity assumption. Dynamic IRL (DIRL) [AJP22] models the reward as a smoothly time-varying combination of spatial goal maps, but its continuity assumption conflicts with evidence that animals switch between *discrete* strategies [ARS+22], and its inference requires T separate Bellman solves, one per timestep. Hierarchical Inverse Q-Learning (HIQL) [ZDLCK+24] replaces smooth variation with a first-order Markov chain over intentions, but the memoryless transition model cannot capture switching driven by cumulative experience such as fatigue or satiation, and its E-step requires the Baum–Welch forward–backward algorithm at $\mathcal{O}(nK^2)$ cost per trajectory. SWIRL [KWW+25] adds state-dependent transition kernels and history-augmented rewards, but represents history via state augmentation: with history length L on $|\mathcal{S}|$ states the augmented state space grows as $|\mathcal{S}|^L$ (e.g., L=2 on 127 states yields 127^2=16,129 augmented states), limiting L to small values in practice. None of these approaches scale naturally beyond small tabular environments.

We propose the *Probabilistic Recurrent Intention Switching Model* (PRISM), a framework that absorbs all temporal complexity into a single lightweight recurrent neural network. At each timestep the network reads an observation from the environment, updates a hidden state that summarizes the trajectory so far, and outputs a soft assignment over a finite set of intentions. This assignment is combined with per-intention Boltzmann policy likelihoods to form a per-step posterior responsibility over intentions. We prove that the resulting expectation-maximization (EM) objective decomposes exactly into independent per-intention reward subproblems, each solvable in closed form via inverse action-value iteration (IAVI) [KHWB20], requiring no variational approximation. The posterior factorizes into independent per-step terms, yielding an $\mathcal{O}(nK)$ E-step. With approximately 50K trainable parameters in a single-layer RNN, PRISM trains in minutes on a laptop GPU and produces human-interpretable reward maps without manual specification of the temporal horizon.

We evaluate PRISM on three domains of increasing complexity, each testing a different property of the framework. Our central application is the *127-node mouse labyrinth* [RZPM21], where PRISM recovers three intentions (water-seeking, homing, exploration) aligned with known biological drives, matching the modes identified by both DIRL [AJP22] and SWIRL [KWW+25] on the same dataset while achieving higher held-out log-likelihood. A *frustration gridworld* with a hidden counter provides controlled validation that PRISM captures provably non-Markovian switching, which is essential for modeling history-dependent behavioral states such as satiation and fatigue. The *BridgeData V2 robotic manipulation dataset* [WBZ+23] tests whether the method generalizes beyond neuroscience: without any supervision, PRISM discovers four temporally coherent manipulation phases (approach, grasp, carry, idle) from human-teleoperated demonstrations. Together, these experiments suggest that discrete intention switching is present in both biological and artificial agents, and that the reward maps PRISM recovers provide a useful lens for interpreting the latent goals behind complex sequential behavior.

In summary, our contributions are: (i) the PRISM framework with a proven EM decomposition and closed-form reward recovery; (ii) an $\mathcal{O}(nK)$ E-step via posterior factorization, compared to the $\mathcal{O}(nK^2)$ forward–backward pass required by Markov-chain-based alternatives; (iii) to our knowledge, the first application of multi-intention IRL to a large-scale robotic manipulation dataset; (iv) recovery of nameable, interpretable intentions across three domains without supervision.

2 Related Work

Inverse reinforcement learning. IRL recovers a reward function from demonstrations, assuming the expert maximizes long-term return. The problem was formalized by [NR+00], who showed it is inherently ill-posed. Maximum entropy IRL [ZMB+08] and its causal variant [ZBD10] resolve this ambiguity via entropy regularization. [KHWB20] derived a closed-form reward solution via inverse action-value iteration (IAVI), which we use as the inner-loop solver in our EM framework. For a comprehensive survey, see [AD21].

Multi-intention and dynamic IRL. Standard IRL assumes a single fixed reward. In practice, agents switch between distinct strategies within an episode [ARS+22]. [BMSL11] cluster entire trajectories by intention but do not allow within-episode switching. Bayesian non-parametric methods [DR11, SS14] avoid fixing the number of intentions but scale poorly. DIRL [AJP22] models reward as a continuously time-varying combination of goal maps; HIQL [ZDLCK+24] uses a first-order Markov chain; SWIRL [KWW+25] adds state-dependent transitions with fixed-window history augmentation. PRISM combines the strengths of all three: discrete switching like HIQL but with memory; history-dependent like SWIRL but learned end-to-end; data-driven like DIRL but with discrete intentions and closed-form reward recovery.

Hierarchical imitation learning and option discovery. The options framework [SPS99] decomposes policies into temporally extended sub-policies. CompILE [KLD+19] segments demonstrations into composable latent skills, and play-data approaches [LKX+20] discover reusable motor primitives from unstructured teleoperation. These methods recover *policies* or *skills*; PRISM instead recovers per-intention *reward functions*, which are more compact, transferable across dynamics, and directly interpretable. PRISM can be seen as performing "inverse option discovery," recovering the reward structure that generates the behavioral options.

Imitation learning at scale. Behavioral cloning [Pom88], GAIL [HE16], and recent generalist robot policies such as RT-1 [BBC+22] learn policies directly from demonstrations without recovering reward structure. PRISM complements these approaches: the intention segmentation it produces could serve as a pre-processing step for skill-conditioned imitation learning.

Interpretable reward and decision models. Reward decomposition [JKF+19] and programmatic RL [VMS+18] seek to make RL decisions transparent. PRISM contributes to this space by recovering discrete, nameable intentions and per-intention reward maps from unlabeled demonstrations.

IRL in robotic manipulation. IRL has been used in autonomous driving [KHWB20] and goal-conditioned manipulation [EZLS22], but always under a single stationary reward. Applying multi-intention IRL to high-dimensional robotic data requires discretizing continuous observations and actions into a finite MDP. We do so by extracting frame-level features from pretrained visual encoders and clustering them via k-means, then compare self-supervised (DINOv2) and vision-language (SigLIP2) representations to study which visual properties matter for reward recovery.

Positioning. PRISM is a table-based, model-based (requires known or estimated transition model P), offline multi-intention IRL method. It operates on fully observed demonstration trajectories: all states, actions, and observations are available before inference begins. Its EM algorithm alternates between closed-form reward recovery via IAVI and gradient-based updates to the recurrent intention network, without end-to-end differentiation through the Bellman equation.

3 Background

3.1 Markov Decision Process

An MDP can be denoted by a tuple $\langle \mathcal{S}, \mathcal{A}, P_a, r, \gamma \rangle$, where $\mathcal{S}$ and $\mathcal{A}$ denote the state- and action-space, respectively; The function $P_a \colon \mathcal{S} \times \mathcal{A} \times \mathcal{S} \to [0,1]$ is the state transition function with $P_a(s,a,s') = \mathbf{P}(s' \mid s,a)$ and $\mathbf{1}^\top P_a(s,a,\cdot) = 1$; The function $r \colon \mathcal{S} \times \mathcal{A} \to \mathbf{R}$ defines the reward function, and $\gamma \in [0,1)$ denotes the discount factor. Additionally, the function $\pi \colon \mathcal{S} \times \mathcal{A} \to [0,1]$ with $\pi(s,a) = \mathbf{P}(a \mid s)$ and $\mathbf{1}^\top \pi(s,\cdot) = 1$ is used to represent the policy according to which actions are selected in the MDP.

3.2 Hidden-intention Markov Decision Process (HI-MDP)

A Hidden-Intention MDP (HI-MDP) extends the standard MDP with a latent intention space $\mathcal{Z}$ and is denoted $\langle \mathcal{Z}, \mathcal{S}, \mathcal{A}, P_a, \{r_z\}_{z \in \mathcal{Z}}, \gamma \rangle$, where the state transition function P_a and policy π are defined as in the standard MDP. At each timestep t, the expert operates under a latent intention $z_t \in \mathcal{Z}$ and receives reward according to $r_{z_t} \colon \mathcal{S} \times \mathcal{A} \to \mathbf{R}$. The expert also perceives an observation $\varphi_t \in \mathbf{R}^m$ from the environment; the observation need not be in one-to-one correspondence with the state and may encode any information available at step t, including the state, the action, or features derived from either. For each trajectory $\xi = \{(s_1,a_1), \dots, (s_n,a_n)\}$ there is a corresponding observation sequence $\psi = \{\varphi_1, \dots, \varphi_n\}$, and we denote the set of all such sequences by $\mathcal{O}$. The mechanism by which observations influence intention assignments is left unspecified in the HI-MDP; PRISM instantiates this mechanism via a learned gating function f_θ (§4.1).

3.3 Inverse Reinforcement Learning

Given the set of expert demonstrations $\mathcal{D}$ in an MDP, where each trajectory $\xi \in \mathcal{D}$ is denoted with a sequence of state-action pairs: $\xi = \{(s_1,a_1), \dots, (s_n,a_n)\}$, the IRL problem consists of determining a reward function r that best explains the observed expert behavior in the form of demonstrated trajectories. Unfortunately, IRL is generally ill-posed because infinitely many reward functions are consistent with the expert's observed behavior. To resolve this issue, various approaches have been proposed (see [AD21] for a detailed review). Particularly, [KHWB20] formulated the IRL problem as a *maximum likelihood estimation* (MLE) problem:

$$
\begin{aligned}
\text{maximize} \quad & \mathop{\mathbf{E}}_{\xi \sim \mathcal{D}} \log \mathbf{P}\left(\xi \mid \pi_r\right) \\
\text{subject to} \quad & \pi_r(s,a) = \exp\Big(Q(s,a) - \log \sum \exp Q(s,\cdot)\Big), \text{for all } s \in \mathcal{S},\ a \in \mathcal{A} \\
& Q(s,a) = r(s,a) + \gamma \sum_{s' \in \mathcal{S}} P(s,a,s') \max_{a' \in \mathcal{A}} Q(s',a'), \text{for all } s \in \mathcal{S},\ a \in \mathcal{A},
\end{aligned}
\tag{1}
$$

where r is the optimization variable and $\mathcal{D}$ is the problem data. By assuming that the expert policy satisfies a Boltzmann distribution, if the environment model P is known, problem (1) can be solved in closed-form via least squares [KHWB20], leading to the model-based *inverse action-value iteration* (IAVI) algorithm. On the other hand, when P is unknown, the problem can be addressed via stochastic approximation, giving rise to the model-free *inverse Q-learning* (IQL) algorithm [KHWB20]. Compared to the popular Maximum Entropy IRL algorithm from [ZMB+08] and some of its variants, the class of IQL algorithms excels at both learning performance and time-efficiency.

4 Methods

4.1 Probabilistic Recurrent Intention Switching Model

We formulate the multi-intention IRL problem under three assumptions.

Assumption 1. *Each expert demonstration step is generated according to a Boltzmann-optimal policy under one of the reward functions in a K-dimensional finite set $\mathcal{R} = \{r_1, \dots, r_K\}$.*

Assumption 2. *Each trajectory ξ in the demonstration set $\mathcal{D}$ is accompanied by an observation sequence $\psi = \{\varphi_1, \dots, \varphi_n\} \in \mathcal{O}$, as defined in §3.2. The observation φ_i may encode any information available at step i, including the state, the action, or features derived from either.*

Assumption 3. *The soft intention assignment at each demonstration step is produced by a parametric function $f_\theta \colon \mathbf{R}^m \to \Delta^K$, where $f_\theta(\varphi_i)_k$ denotes the weight assigned to intention k given observation φ_i, for $k = 1, \dots, K$. The function f_θ may maintain internal state across steps within a trajectory, as realized by recurrent neural networks.*

Under these assumptions the expert's decision process is fully specified by $\Theta = \{\theta, \mathcal{R}\}$. We refer to the resulting framework as the *Probabilistic Recurrent Intention Switching Model* (PRISM). The PRISM inference problem consists of determining (1) a set of reward functions and (2) the intention index for each demonstration step that best jointly explain the observed expert behavior. An expectation-maximization (EM) algorithm can be devised to iteratively learn Θ. For convenience, we introduce $\eta = \{z_1, \dots, z_n\}$ as the predicted sequence of intention indices for trajectory ξ and corresponding observations ψ. Then each iteration of the EM process maximizes the auxiliary function:

$$\text{maximize} \quad J(\Theta^+ \mid \Theta) = \mathop{\mathbf{E}}_{(\xi,\psi)\sim(\mathcal{D},\mathcal{O}),\,\eta} \log \mathbf{P}(\xi, \eta \mid \psi, \Theta^+), \tag{2}$$

where Θ^+ is the optimization variable and $(\mathcal{D}, \mathcal{O}), \Theta$ are the problem data.

Theorem 1. *Solving problem (2) is equivalent to solving a sequence of independent optimization problems: first maximize over the intention network parameters θ^+,*

$$\text{maximize (over } \theta^+) \quad \mathbf{E}_{(\xi,\psi)\sim(\mathcal{D},\mathcal{O})} \left(\textstyle\sum_{k=1}^{K} \sum_{i=1}^{n} \mathbf{P}(z_i{=}k \mid \xi, \psi, \Theta) \log f_{\theta^+}(\varphi_i)_k \right), \tag{3}$$

and then maximize independently over each reward $r_k^+ \in \mathcal{R}^+$,

$$\begin{array}{ll} \text{maximize (over } r_k^+) & \mathbf{E}_{(\xi,\psi)\sim(\mathcal{D},\mathcal{O})} \left(\sum_{i=1}^{n} \mathbf{P}(z_i{=}k \mid \xi, \psi, \Theta) \log \pi_{r_k^+}(a_i \mid s_i) \right) \\ \text{subject to} & \pi_{r_k^+}(s,a) = \exp\big(Q(s,a) - \log \sum_{a' \in \mathcal{A}} \exp Q(s,a')\big) \\ & Q(s,a) = r_k^+(s,a) + \gamma \sum_{s'} P(s' \mid s,a) \max_{a' \in \mathcal{A}} Q(s',a') \\ & s \in \mathcal{S}, \quad a \in \mathcal{A}. \end{array} \tag{4}$$

This decomposition is exact, requiring no variational approximation. Each reward subproblem (4) reduces to a weighted IAVI problem solvable in closed form. Detailed proof is given in § A.1.

Posterior factorization and complexity. A key structural consequence is that, conditioned on (ξ, ψ, Θ), the intention variables $\{z_1, \ldots, z_n\}$ are mutually independent. The posterior decomposes as $\mathbf{P}(\eta \mid \xi, \psi, \Theta) = \prod_{i=1}^{n} \mathbf{P}(z_i \mid \xi, \psi, \Theta)$, where each per-step responsibility is the normalized product of the gating output and the per-intention policy:

$$P(z_i{=}k \mid \xi, \psi, \Theta) = \frac{f_\theta(\varphi_i)_k \, \pi_{r_k}(a_i \mid s_i)}{\sum_{j=1}^{K} f_\theta(\varphi_i)_j \, \pi_{r_j}(a_i \mid s_i)}, \tag{5}$$

where f_θ assigns soft responsibility over intentions given the observed context φ_i, and π_{r_k} scores how well the observed action a_i is explained under intention k. Since the intention variables decouple across time steps, the E-step requires only $\mathcal{O}(nK)$ evaluations per trajectory. This stands in contrast to HIQL [ZDLCK+24], where a Markov transition matrix couples consecutive intentions (z_{t-1}, z_t) and the E-step requires the Baum–Welch forward–backward algorithm at $\mathcal{O}(nK^2)$ cost per trajectory. The recurrent network thus serves a dual role: it captures arbitrarily long-range history dependence that a first-order Markov chain cannot represent, while simultaneously simplifying inference.

4.2 Training Objective

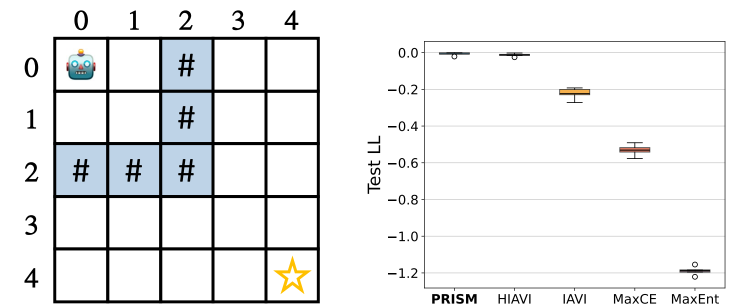
The training objective for f_θ combines the negative log-likelihood from the M-step with temporal smoothness penalties. Writing $w_{i,k}$ for the per-step responsibility in Eq. (5):

$$\mathcal{L} \;=\; \mathcal{L}_{\mathrm{NLL}} \;+\; \lambda_{\ell_1} \mathcal{L}_{\ell_1} \;+\; \lambda_{\mathrm{kl}} \mathcal{L}_{\mathrm{kl}}, \tag{6}$$

where

$$\begin{aligned} \mathcal{L}_{\mathrm{NLL}} &= - \mathop{\mathbf{E}}_{(\xi,\psi)\sim(\mathcal{D},\mathcal{O})} \sum_{i=1}^{n} \sum_{k=1}^{K} w_{i,k} \log f_\theta(\varphi_i)_k, \\ \mathcal{L}_{\ell_1} &= \mathop{\mathbf{E}}_{(\xi,\psi)\sim(\mathcal{D},\mathcal{O})} \sum_{i=2}^{n} \sum_{k=1}^{K} w_{i,k} \big| f_\theta(\varphi_i)_k - f_\theta(\varphi_{i-1})_k \big|, \\ \mathcal{L}_{\mathrm{kl}} &= \mathop{\mathbf{E}}_{(\xi,\psi)\sim(\mathcal{D},\mathcal{O})} \sum_{i=2}^{n} D_{\mathrm{kl}}\big(f_\theta(\varphi_{i-1}) \,\|\, f_\theta(\varphi_i)\big). \end{aligned}$$

The smoothness terms are computed over consecutive time steps within each trajectory. The ℓ_1-penalty directly suppresses absolute changes in intention probabilities between consecutive steps, providing strong compression of rapid switches; however, used alone it can



(a) The gridworld environment. (b) Log-likelihood on test dataset.

Figure 1 Frustration gridworld.

over-smooth genuine transitions and reduce likelihood. The KL-divergence term penalizes distributional shift more gently, preserving the shape of the intention distribution but offering weaker suppression on its own. The two are complementary: ℓ_1 provides sparse switching while KL preserves smooth transitions. Both weights $\lambda_{\ell_1}, \lambda_{\mathrm{kl}} \geq 0$ are set in §5.2. We note that these penalties act as discriminative regularizers rather than a probabilistic prior over intention sequences. The RNN hidden state provides implicit temporal modeling, while the regularizers encourage the output distribution to vary smoothly.

4.3 Intention Network Architecture

The intention network maps a sequence of observations $(\varphi_1, \ldots, \varphi_n)$ to a per-step distribution over K intentions. Each observation φ_t is embedded into a d-dimensional vector; the sequence of embeddings is processed by a recurrent encoder (RNN, LSTM, or Transformer); and each encoder output is projected to K logits followed by a softmax to produce $f_\theta(\varphi_t) \in \Delta^K$. In the tabular experiments we set $\varphi_t = (s_t, a_t)$ and use learned state and action embedding matrices; in visual domains φ_t may be the output of a pretrained encoder. A single-layer IntentionRNN with d=128 and K=4 has approximately 50K trainable parameters.

5 Experiments

5.1 Gridworld

5.1.1 Environment setup

The simulated gridworld environment features a 5×5 map, where the agent chooses from five actions at each step: up, down, left, right, or remain stationary. The agent starts from $(0, 0)$ and operates under stochastic dynamics such that intended actions succeed with 90% probability, with the remaining 10% causing random movement. It has two possible policies: π_{goal} prefers going to the destination $(4, 4)$, and π_{abandon} prefers returning to origin $(0, 0)$. Each episode starts with the π_{goal} policy.

The key feature of this environment is a hidden frustration mechanism that makes intention switching non-Markovian. The expert maintains a counter c, initially zero, that increments each time it enters a barrier state $\#$. At each barrier encounter, the probability of switching to the other policy is $\min\{0.15c, 0.9\}$. Early encounters rarely trigger a switch, but repeated encounters make switching increasingly likely. The counter resets upon switching. Each episode ends when the agent reaches either the origin or the destination.

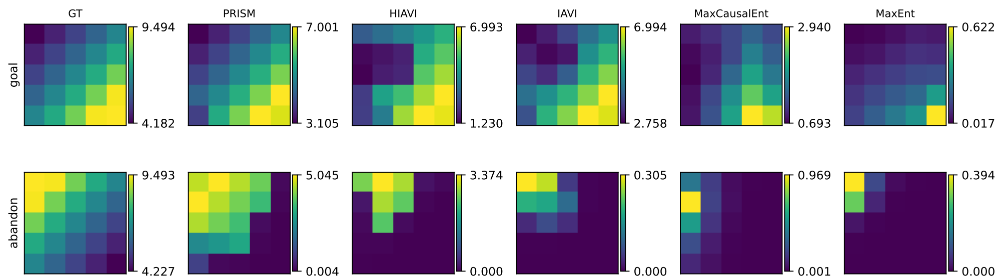


Figure 2 State-value heatmaps on the frustration gridworld.

5.1.2 Results

We compared PRISM against four baselines: HIQL [ZDLCK+24] ($K = 2$), its single-intention counterpart IAVI [KHWB20], maximum causal entropy IRL [ZBD10], and maximum entropy IRL [ZMB+08]. The dataset consisted of 1024 trajectories evaluated with 5-fold cross-validation.

Quantitative results. PRISM achieves the highest test log-likelihood with the smallest variance across folds (Figure 1b), and the lowest EVD under both intentions (Table 3). HIQL follows closely in log-likelihood but shows higher EVD under the abandon intention, attributable to its Markov model being unable to capture the cumulative counter. The single-intention baselines perform substantially worse on both metrics, confirming that modeling multiple intentions is necessary in this environment.

Qualitative results. The state-value heatmaps (Figure 2) tell a consistent story. PRISM's recovered values closely match the ground truth under both intentions: the goal-value gradient increases toward $(4, 4)$ while attenuating in the barrier-dense region, and the abandon-value concentrates near the origin. HIQL recovers a reasonable goal heatmap but a notably weaker abandon heatmap. The single-intention baselines produce spatially smooth maps that fail to distinguish the two modes. The recovered reward maps and greedy actions (Figure 3) show the same pattern: PRISM's per-state arrows align with the expert's strategies under both intentions, whereas HIQL shows misaligned actions under the abandon policy and the single-intention baselines produce diffuse or conflicting directional signals.

5.2 Mouse Labyrinth Navigation

5.2.1 Dataset and benchmarks

We evaluate PRISM on the freely moving mouse navigation dataset from [RZPM21], the same dataset used in [KWW+25] and [ZDLCK+24]. Ten water-restricted mice explored a 127-node labyrinth for 7 hours in darkness; a water reward was available at a designated

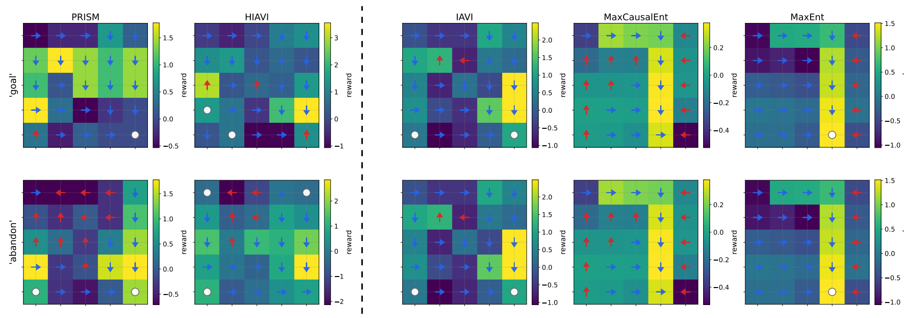


Figure 3 Recovered per-state reward maps and greedy actions on the frustration gridworld.

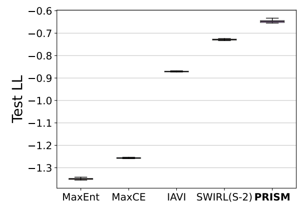
end node but could be collected at most once every 90 seconds. This 90-second constraint forces mice to leave the port after drinking, creating a non-Markovian reward structure in which the value of returning to the water node depends on elapsed time rather than the current state alone.

Following [KWW+25], we segment the raw node visit sequences into 238 trajectories of 500 time steps each, retaining the full behavioral detail without stereotyped-path clustering. Models are trained on 80% of trajectories and evaluated on the held-out 20% using 5-fold cross-validation, with held-out negative log-likelihood (NLL) as the primary metric. We compare PRISM ($K = 3$, IntentionRNN) against three baselines: IAVI (single intention), SWIRL (S-2) from [KWW+25] which incorporates state-dependent transition kernels and history-dependent rewards, and maximum causal entropy IRL [ZBD10]. For this experiment we activate both smoothness penalties ($\lambda_{\ell_1} = 2.22$, $\lambda_{\mathrm{kl}} = 1.48$); in the gridworld and Bridge V2 experiments both are set to zero.

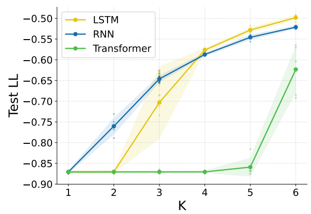
5.2.2 Behavior prediction performance

Figure 4a shows the distribution of test NLL across folds. PRISM achieves the highest test NLL (-0.65), outperforming SWIRL S-2 (-0.73), the single-intention baseline (IAVI), and maximum causal entropy IRL. The improvement over SWIRL is notable because both methods model history-dependent intention dynamics on the same data: PRISM obtains a better fit while learning the relevant history horizon end-to-end, without the manual state augmentation that SWIRL requires.

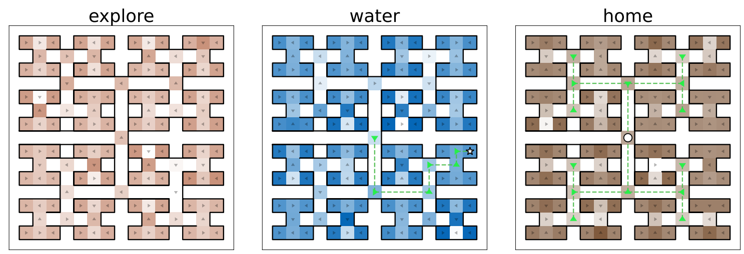
Figure 4b shows test log-likelihood as a function of the number of latent intentions K for three intention network architectures. All three improve monotonically with K, confirming the presence of multiple behavioral modes; we adopt the vanilla RNN as default for its simpler dynamics and stable performance. We select $K{=}3$ because the labyrinth task contains only two explicit behavioral events (water-seeking and homing), and the plateau in test log-likelihood beginning at $K{=}4$ (Figure 4b) indicates that three intentions already capture the dominant behavioral structure, with the third corresponding to exploration. This setting also allows direct comparison with SWIRL [KWW+25], which reports its primary results at $K{=}3$. The RNN's simpler hidden dynamics also enable the interpretability analysis in §C; all main conclusions hold under LSTM as well. Exact numerical values are provided in Table 5 (Appendix).



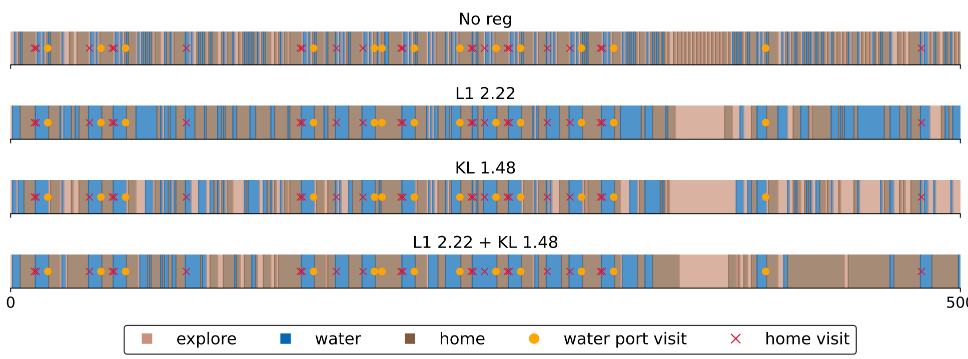
(a) Test log-likelihood across baselines.



(b) Test log-likelihood for three intention temporal models.



(c) Recovered reward maps and greedy policy flow for the three inferred intentions (star: water port; circle: entrance).



(d)

Figure 4 Labyrinth results (PRISM, K=3, IntentionRNN, hybrid regularization, 238 mouse trajectories). **(a)** Test log-likelihood: PRISM outperforms all baselines. **(b)** Test log-likelihood vs K for three architectures. **(c)** Recovered reward maps and greedy policy flow for the three inferred intentions (star: water port; circle: entrance). **(d)** Temporal intention segmentation under four regularization configurations. Orange dots: water-port visits; red crosses: home visits.

5.2.3 Regularization and intention segmentation

All results in Figure 4a use the hybrid regularization described above. We now examine how different regularization settings affect the recovered reward maps and intention segments.

Figure 4c shows the greedy policy and per-state action confidence for each intention, where color depth indicates how decisively the greedy action dominates over alternatives. Under *water*, the greedy policy traces a clear path from the entrance toward the water port, with high confidence along the main corridor. Under *home*, the major branching nodes consistently point toward the entrance, forming convergent flow inward. Under *explore*, actions are distributed across peripheral nodes with no dominant target and lower overall confidence, reflecting the diffuse nature of exploratory behavior. These three modes are consistent with those identified by SWIRL [KWW+25].

Figure 4d shows per-step intention assignments for a representative held-out trajectory under four regularization configurations, spanning the regularization spectrum from unconstrained to fully penalized. Without regularization, the model produces rapid switching among all three intentions. Both ℓ_1-regularizer and KL-divergence regularizer reduce switching compared to the unregularized case: The ℓ_1-regularizer produces the sparsest segments due to its constant-magnitude gradient penalty, while KL-divergence regularizer suppresses large distributional jumps but tolerates gradual shifts, resulting in more frequent low-amplitude transitions. The hybrid settings inherit the temporal sparsity induced by the ℓ_1-norm while preserving the smooth transition behavior from the KL-divergence, and their intention boundaries align well with behaviorally meaningful events — water-port visits (orange dots) and home visits (red crosses). The full-weight hybrid ($\lambda_{\ell_1} = 2.22$, $\lambda_{\text{kl}} = 1.48$) is used as the default setting for results reported in Figure 4a.

Table 1 reports wall-clock timing. PRISM converges in $\sim$140 EM iterations on the labyrinth ($\sim$5 min total) on a NVIDIA RTX 3060 mobile GPU, and $\sim$80 iterations on Bridge V2 ($\sim$12 min), on a NVIDIA RTX 5060Ti GPU.

Dataset	**E-step**	**M-step (per latent)**		**Total / iter**	**Conv. iters**
	(posterior)	IAVI	Intention net		
Labyrinth	0.03 s	0.60 s	0.09 s	1.92 s	$\sim$140
Bridge V2	4.10 s	1.53 s	2.90 s	8.54 s	$\sim$80

Table 1 Wall-clock time per EM iteration.

5.3 BridgeData V2: Robotic Manipulation

In our third experiment, we push PRISM to a qualitatively new regime with high-dimensional visual observations and no prior knowledge of intentions, which constitutes, to our knowledge, the first application of multi-intention IRL to a large-scale robotic manipulation dataset. We apply PRISM to the BridgeData V2 robot arm dataset from [WBZ+23], which presents high-dimensional continuous observations (256×256 RGB) and actions (7D end-effector motion plus gripper) that must be discretized before tabular IRL can be applied.

This experiment serves two purposes. First, it tests whether PRISM scales to substantially larger state and action spaces than the previous settings. Second, it investigates whether intention structure can be recovered from robot demonstrations without explicit intention labels. In the labyrinth experiment, the recovered intentions (water-seeking, home-seeking, exploration) align with known biological drives of water-deprived mice. BridgeData V2 contains no such prior knowledge: each trajectory is a human-teleoperated manipulation sequence annotated only with a prompt description of the task. If PRISM can nonetheless decompose these trajectories into temporally coherent segments that correspond to recognizable manipulation phases, it provides evidence that intention structure is a general property of goal-directed sequential behavior, not specific to biological agents.

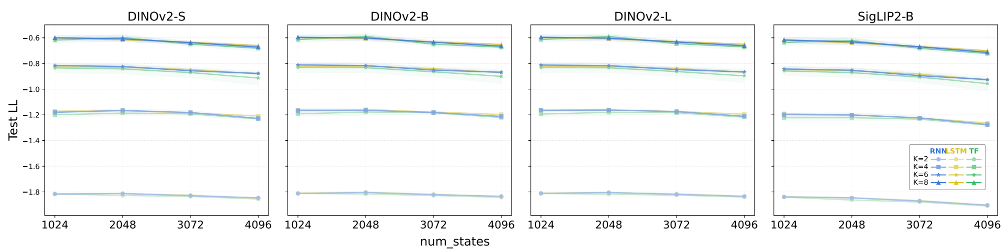


Figure 5 BridgeData V2: test log-likelihood by encoder, intention network, and number of latents K.

5.3.1 Continuous-to-discrete pipeline

PRISM operates on tabular MDPs with discrete states and actions. To bridge the gap between the raw continuous modalities in BridgeData V2 and this requirement, we construct a two-stage discretization pipeline. In the first stage, each 256×256 RGB frame is forwarded through a frozen pretrained visual encoder to obtain a fixed-dimensional embedding. For actions, each 7-dimensional continuous vector (6D delta end-effector pose plus gripper command) is retained in its raw form. In the second stage, k-means clustering is applied independently to the pooled state embeddings and action vectors across the full dataset, producing $|\mathcal{S}|$ discrete state tokens and $|\mathcal{A}|$ discrete action tokens. Each trajectory is then re-encoded as a sequence of integer state–action pairs, yielding a discrete MDP that is directly amenable to PRISM.

The quality of the discrete state representation depends on the visual encoder. We compare two families of pretrained encoders that differ in their training objective. The first family is DINOv2 [ODM+23], a self-supervised vision encoder trained to produce general-purpose visual features without any language or task supervision. We test three model scales: ViT-S/14, ViT-B/14, and ViT-L/14. The second is SigLIP2 [TGW+25], a vision-language encoder trained with a sigmoid pairwise contrastive loss to align image representations with natural language descriptions. We test its ViT-B scale. The two training objectives lead to different discretization behaviors. DINOv2 preserves fine-grained visual distinctions (gripper pose, object orientation, spatial layout), producing clusters that are visually homogeneous but spread across a larger portion of the codebook. SigLIP2 aligns its representations with semantic content, collapsing visually distinct frames that share the same language description into fewer, more heavily visited clusters. Which property matters more for IRL is not obvious: denser clusters provide more reliable transition estimates for IAVI, but overly coarse clusters may alias states that require different actions, degrading the recovered policy. We quantify this trade-off in Table 2 and evaluate its downstream effect on intention recovery in the results below.

Figure 6 Per-timestep intention assignments on BridgeData V2 trajectories. Frame borders are color-coded by predicted intention: orange (CARRY), blue (IDLE), green (GRASP), magenta (APPROACH/DEPART).

5.3.2 Discretization granularity

The resolution of the k-means codebook controls a trade-off between state aliasing and transition sparsity: a larger codebook $|\mathcal{S}|$ assigns finer-grained cluster identities but reduces the number of revisits per state, degrading the empirical transition probability estimates that IAVI relies on. We ablate over $|\mathcal{S}| \in \{1024, 2048, 3072, 4096\}$ with a fixed action vocabulary $|\mathcal{A}| = 32$ for all four encoder variants (Table 2).

Despite its sparser transition tables, DINOv2 achieves higher held-out test log-likelihood than SigLIP2 (Figure 5). This result reveals that for IRL, the action-level distinguishability of states matters more than the density of transition estimates. SigLIP2's semantic alignment collapses frames that look different but share a language description into the same cluster. When those frames in fact require different actions (e.g., approaching an object from the left versus the right), the resulting state aliases degrade the per-state policy that IAVI recovers. DINOv2 retains these visual distinctions at the cost of sparser tables, but the finer-grained states better support accurate reward and policy recovery.

We adopt $|\mathcal{S}| = 2048$ with DINOv2-S as the default configuration: discretization quality is governed by encoder architecture rather than scale, with DINOv2-S, -B, and -L producing nearly identical statistics, so we adopt the smallest variant to minimize compute. The critical gap is between DINOv2 and SigLIP2: at $|\mathcal{S}|$=2048, SigLIP2-B collapses visually distinct frames into shared tokens (11.8 average revisits vs 7.1 for DINOv2-S), reducing the action-level state discrimination that IRL requires. At our default granularity ($|\mathcal{S}|$=2048, DINOv2-S), approximately 75% of visited states receive multiple observations, providing adequate support for transition estimation. This encoder advantage is reflected in test log-likelihood: DINOv2 consistently outperforms SigLIP2 across all configurations (Figure 5).

Table 2 Effect of k-means state granularity on trajectory discretization statistics (mean±std across trajectories). **Coverage**: fraction of all clusters visited per trajectory (lower = finer discretization). **Avg. revisits**: mean number of time steps each visited state is revisited per trajectory (higher = denser coverage). **Singleton**: fraction of visited states seen only once per trajectory (higher = more "passing-through", fewer revisits).

Encoder	**Metric**	**Number of states** $\|\mathcal{S}\|$			
		1024	2048	3072	4096
DINOv2-S	Coverage (%)	0.64±0.40	0.37±0.23	0.26±0.17	0.21±0.13
	Avg. revisits	8.0±6.1	7.1±5.7	6.8±5.6	6.5±5.6
	Singleton (%)	21.7±19.1	25.5±19.9	27.6±20.5	28.9±20.9
DINOv2-B	Coverage (%)	0.69±0.42	0.40±0.25	0.28±0.18	0.22±0.14
	Avg. revisits	7.6±5.9	6.6±5.4	6.3±5.3	6.0±5.1
	Singleton (%)	23.4±19.3	27.4±20.2	29.6±20.7	30.9±21.0
DINOv2-L	Coverage (%)	0.69±0.42	0.40±0.25	0.29±0.18	0.23±0.14
	Avg. revisits	7.4±5.9	6.5±5.4	6.2±5.3	5.9±5.2
	Singleton (%)	23.4±19.1	27.1±20.0	29.2±20.4	30.8±20.7
SigLIP2-B	Coverage (%)	0.38±0.25	0.22±0.15	0.16±0.11	0.13±0.09
	Avg. revisits	13.3±9.6	11.8±8.9	11.0±8.4	10.5±8.1
	Singleton (%)	14.7±18.5	17.5±19.2	19.0±19.6	20.0±19.9

5.3.3 Results

Figure 5 reports held-out test log-likelihood as a function of $|\mathcal{S}|$, encoder, number of latent intentions K, and intention network architecture. Test log-likelihood improves monotonically with K (Figure 5): from -1.81 at K=2 to -0.82 at K=6 with no saturation, unlike the labyrinth where gains plateau by K=4. IntentionRNN and IntentionLSTM perform similarly; IntentionTransformer lags at small K but narrows the gap as K increases, reproducing the same ranking observed on the labyrinth and supporting IntentionRNN as a robust default. We report visualizations at K=4 because Bridge V2 trajectories are relatively short; at K=5 and beyond, segments become excessively brief and lose interpretability, with new classes capturing transient gripper adjustments rather than distinct behavioral modes.

Figure 6 visualizes per-timestep intention assignments (K=4, DINOv2-S). Without any supervision, PRISM recovers four classes: APPROACH/DEPART (arm moving toward or away from target), GRASP (gripper closing), CARRY (transporting object), and IDLE (minor adjustments). The boundaries are sharp and consistent across trajectories with different objects and environments, confirming temporally coherent segmentation, though we note that no ground-truth intention labels exist for formal validation.

6 Conclusion

We presented PRISM, an EM-based framework for multi-intention IRL that parameterizes intention dynamics through a lightweight recurrent neural network. By replacing the Markov chain of HIQL and the fixed-window augmentation of SWIRL with an end-to-end learned recurrent mapping, PRISM adapts to the temporal structure of each dataset without manual design choices, while retaining closed-form reward recovery. The proven EM decomposition separates the reward and gating subproblems exactly, and the $\mathcal{O}(nK)$ E-step scales gracefully. Experiments on three progressively complex domains, from an abstract gridworld validating non-Markovian theory, through real mouse behavior confirming biological interpretability, to large-scale robotic manipulation demonstrating cross-domain generality, show that PRISM consistently achieves the highest held-out likelihood while recovering nameable intentions from unlabeled data. The recovered reward maps provide a structured characterization of the latent goals behind observed behavior, revealing not just *what* an agent did but *which objective* it was pursuing at each moment.

7 Discussion and Outlook

(a) Tabular assumption. PRISM currently requires a discrete MDP. Our k-means pipeline bridges this gap for visual domains but introduces quantization error. The k-means codebook optimizes within-cluster Euclidean variance, which has no guaranteed relationship to the behavioral equivalence required for correct reward recovery in IRL. Learned codebooks such as VQ-BeT [LWE+24] or SAQ [LDW+23] could adapt cluster boundaries to maximize reward-recovery quality. However, discretization has fundamental limits: even cutting-edge vision-language-action models such as $\pi_{0.5}$ [IBB+25] employ action tokenizers like FAST for pre-training their language-model backbone, yet such tokenization serves the model architecture rather than enabling fine-grained dexterous control directly. The continuous structure that dexterous manipulation demands is inherently lost through any discretization, which motivates moving beyond tabular methods entirely.

(b) Model-free extension. The natural path forward is to replace IAVI with model-free inverse Q-learning [KHWB20], which operates directly in continuous state–action spaces via function approximation and does not require an explicit transition matrix. This would remove both the discretization bottleneck of (a) and the model-based dependency, enabling PRISM to scale to dexterous robotic domains where tabular representations are infeasible.

(c) Online extension. PRISM operates offline over fully observed trajectories, which suffices for post-hoc behavioral analysis but precludes real-time intention monitoring. Introducing a probabilistic transition prior $P(z_t \mid z_{t-1}, \varphi_{t-1})$ would enable sequential filtering for online applications. In preliminary experiments, such a Bayesian variant achieved comparable log-likelihood but produced less interpretable reward maps, suggesting that the prior structure requires careful design to preserve reward quality.

(d) Model selection. Test log-likelihood improves monotonically with K, but beyond the true number of behavioral modes the additional intentions fragment trajectories into unnameable segments, particularly when trajectories are short (§5.3). A single-layer RNN with 128 hidden units already saturates performance on both datasets (Table 6), suggesting that the switching dynamics in these domains are low-dimensional. Longer, more complex demonstrations would be needed to justify both larger K and more expressive sequence models.

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A Theoretical and technical details

A.1 Proof of Theorem 1

Proof. The objective function $J(\Theta^+ \mid \Theta)$ from problem (2) can be written as:

$$
\begin{aligned}
& J(\Theta^+ \mid \Theta) \qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad (A.1) \\
= \; & \mathop{\mathbf{E}}_{(\xi,\psi)\sim(\mathcal{D},\mathcal{O}),\,\eta} \log \mathbf{P}(\xi, \eta \mid \psi, \Theta^+) \\
= \; & \mathop{\mathbf{E}}_{(\xi,\psi)\sim(\mathcal{D},\mathcal{O})} \left(\sum_{\eta} \mathbf{P}(\eta \mid \xi, \psi, \Theta) \; \log \mathbf{P}(\xi, \eta \mid \psi, \Theta^+) \right) \\
= \; & \mathop{\mathbf{E}}_{(\xi,\psi)\sim(\mathcal{D},\mathcal{O})} \left(\sum_{\eta} \mathbf{P}(\eta \mid \xi, \psi, \Theta) \; \sum_{i=1}^{n} \log \mathbf{P}(z_i^+{=}z_i \mid \varphi_i, \theta^+) \; \mathbf{P}\big((s_i, a_i) \mid r_{z_i}^+\big) \right) \\
= \; & \mathop{\mathbf{E}}_{(\xi,\psi)\sim(\mathcal{D},\mathcal{O})} \left(\sum_{\eta} \mathbf{P}(\eta \mid \xi, \psi, \Theta) \; \sum_{i=1}^{n} \sum_{k=1}^{K} \mathbb{I}_k(z_i) \; \log \mathbf{P}(z_i^+{=}k \mid \varphi_i, \theta^+) \; \mathbf{P}\big((s_i, a_i) \mid r_k^+\big) \right) \\
= \; & \mathop{\mathbf{E}}_{(\xi,\psi)\sim(\mathcal{D},\mathcal{O})} \left(\sum_{k=1}^{K} \sum_{i=1}^{n} \underbrace{\sum_{\eta} \mathbf{P}(\eta \mid \xi, \psi, \Theta)\, \mathbb{I}_k(z_i)}_{=\; \mathbf{P}(z_i \,=\, k \mid \xi, \psi, \Theta)} \log \mathbf{P}(z_i^+{=}k \mid \varphi_i, \theta^+) \; \mathbf{P}\big((s_i, a_i) \mid r_k^+\big) \right) \\
= \; & \underbrace{\mathop{\mathbf{E}}_{(\xi,\psi)\sim(\mathcal{D},\mathcal{O})} \left(\sum_{k=1}^{K} \sum_{i=1}^{n} \mathbf{P}(z_i{=}k \mid \xi, \psi, \Theta) \; \log \mathbf{P}(z_i^+{=}k \mid \varphi_i, \theta^+) \right)}_{\text{(I): depends only on } \theta^+} \qquad\qquad\qquad\qquad (A.2) \\
& \quad + \underbrace{\mathop{\mathbf{E}}_{(\xi,\psi)\sim(\mathcal{D},\mathcal{O})} \left(\sum_{k=1}^{K} \sum_{i=1}^{n} \mathbf{P}(z_i{=}k \mid \xi, \psi, \Theta) \; \log \mathbf{P}\big((s_i, a_i) \mid r_k^+\big) \right)}_{\text{(II): depends only on } \mathcal{R}^+}, \qquad\qquad\qquad (A.3)
\end{aligned}
$$

where $\mathbb{I}_k$ is the indicator function with $\mathbb{I}_k(x) = 1$ for $x = k$ and 0 otherwise. Thus maximizing $J(\Theta^+ \mid \Theta)$ over Θ^+ is equivalent to separately maximizing (A.2) over θ^+ and (A.3) over $\mathcal{R}^+ = \{r_1^+, \ldots, r_K^+\}$.

By Assumption 3, $f_\theta(\varphi)_k = \mathbf{P}(r_k \mid \varphi)$, so the first optimization problem becomes (3). In (A.3), distinct k share no parameters, so the maximization decomposes into K independent subproblems. Each has the same structure as the single-intention IRL problem (1), with the i-th demonstration weighted by $\mathbf{P}(z_i = k \mid \xi, \psi, \Theta)$, yielding (4). The Boltzmann-policy constraints are introduced to make each subproblem tractable via IAVI.

A.2 Intention Network Architecture Details

We use a single intention network that can be instantiated as an RNN, LSTM, or Transformer. In all cases, the network takes a sequence of trajectory features and outputs, at each time step, a distribution over K intention classes. Each step (s_t, a_t) is encoded by two independent learned embedding matrices $E_s \in \mathbf{R}^{|\mathcal{S}|\times d}$ and $E_a \in \mathbf{R}^{|\mathcal{A}|\times d}$, producing a d-dimensional input vector $x_t = E_s(s_t) + E_a(a_t)$. Batches of variable-length trajectories are zero-padded to a common length, and a binary mask indicates valid time steps. The recurrent variants (RNN, LSTM) pass the sequence of embeddings through a single recurrent layer with hidden dimension d', using the padding mask to pack and unpack sequences during computation, and project each hidden state to K logits via a linear layer. The Transformer variant additionally applies sinusoidal positional encoding to x_t before passing the sequence through a standard encoder with a src-key-padding mask. In all variants, the K logits are converted to an intention distribution $f_\theta(\varphi_t) \in \Delta^K$ via a softmax, which is then combined with the K agents' policy log-probabilities and normalized according to Equation (5) to obtain the per-step posterior over intentions. This posterior serves as the soft assignment in the E-step of the EM algorithm.

A.3 Gridworld

Method	**EVD (MAE)**		**EVD (s_0)**	
	goal	abandon	goal	abandon
PRISM ($K = 2$)	$\mathbf{2.40 \pm 0.37}$	$\mathbf{5.60 \pm 0.45}$	$\mathbf{-1.74 \pm 0.59}$	$\mathbf{-6.02 \pm 0.85}$
HIQL ($K = 2$)	2.51 ± 0.13	6.12 ± 0.38	-1.95 ± 0.54	-6.80 ± 1.25
IAVI ($K = 1$)	2.06 ± 0.02	6.74 ± 0.00	-1.41 ± 0.01	-9.20 ± 0.00
MaxCausalEnt ($K = 1$)	5.32 ± 0.00	6.67 ± 0.00	-3.32 ± 0.00	-9.12 ± 0.00
MaxEnt ($K = 1$)	6.61 ± 0.00	6.76 ± 0.00	-4.16 ± 0.00	-9.10 ± 0.00

Table 3 Expected value difference on the frustration gridworld (mean ± std, 5-fold cross-validation). EVD (MAE) is the mean absolute error between the state-value function under the true reward for the expert policy and that for the Boltzmann-optimal policy w.r.t. the learnt reward, averaged over all states. EVD (s_0) reports the same quantity evaluated at the initial state $(0, 0)$. For multi-intention methods ($K = 2$), each inferred intention is assigned to the best-matching ground-truth intention; for single-intention methods ($K = 1$), the same learnt policy is evaluated under both true reward functions. Closer to zero is better.

A.4 Default hyperparameters

Table 4 lists the primary configuration used in our main experiments.

B Ablation Experiments

B.1 Temporal neural network

C Interpretable RNN

A natural question is whether the trained RNN's internal dynamics reflect the discrete intention structure that PRISM recovers. We investigate this by projecting the hidden states onto their principal components and examining the geometry of the resulting trajectories.

We collect 95,000 hidden states across 190 labyrinth training trajectories (128-D each) and project them via PCA. The top three components capture approximately 95% of the

Table 4 Default hyperparameters for the labyrinth and Bridge V2 experiments unless otherwise noted.

Category	**Hyperparameter**	**Symbol**	**Labyrinth**	**Bridge V2**
MDP	Num. states	$\lvert\mathcal{S}\rvert$	127	2048
	Num. actions	$\lvert\mathcal{A}\rvert$	4	32
	Discount factor	γ	0.97	0.97
EM	Max EM iterations		180	150
	Random seed		42	42
Intention model	Num. latent intentions	K	3	4
	Architecture		IntentionRNN	IntentionRNN
Intention net	Embedding dim.	d	128	128
	RNN / LSTM hidden dim.	d'	128	128
	Num. layers		1	1
	Attn. heads (Transformer)		4	4
Optimiser	Learning rate		10^{-3}	10^{-3}
	RNN / LSTM epochs per M-step		1	1
	Transformer epochs per M-step		8	8
Regularisation	Regularisation type		KL + L1	
	L1 smoothness weight	λ_{ℓ_1}	2.22	0.0
	KL smoothness weight	λ_{kl}	1.48	0.0

K	**Labyrinth**		**Bridge V2**	
	Train LL	Test LL	Train LL	Test LL
1	-0.86801	-0.87071	-2.45171	-2.52187
2	-0.75432	-0.76024	-1.74644	-1.81282
3	-0.63754	-0.64624	-1.36307	-1.43000
4	-0.58141	-0.58714	-1.09903	-1.16584
5	-0.53726	-0.54573	-0.90255	-0.96912
6	-0.51220	-0.52129	-0.75780	-0.82373

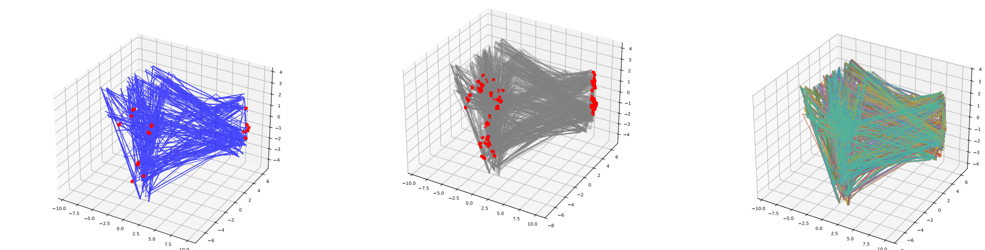
Table 5 Log-likelihood vs number of latent intentions K (IntentionRNN, 5-fold CV, 3 random seeds).

total variance (82.5%, 9.0%, 3.3%), indicating that the RNN operates on a roughly three-dimensional manifold that is consistent with the $K = 3$ intention structure, where three degrees of freedom suffice to encode which intention is active and how far it has progressed.

Projected trajectories trace smooth, distinct paths through 3D PCA space (Figures C.1a and C.1c), confirming that the hidden state encodes trajectory-specific context rather than collapsing onto a single template. We identify fixed-point candidates as timesteps where $\|h_t - h_{t-1}\|$ falls below the 5th percentile. These near-stationary states cluster in a small number of spatially separated regions (Figure C.1b), suggestive of attractor-like dynamics during periods of sustained intention. Whether these clusters align with the three inferred intentions, and what behavioral covariate the dominant first PC encodes, remain open ques-

Model	**Labyrinth**		**Bridge V2**	
	Train LL	Test LL	Train LL	Test LL
IntentionRNN	-0.63754	-0.64624	-1.09903	-1.16584
IntentionLSTM	-0.60993	-0.62705	-1.10082	-1.16839
IntentionTransformer	-0.86797	-0.87067	-1.12025	-1.18638

Table 6 Log-likelihood by intention network architecture (d=128, single layer, 5-fold CV).



(a) Single trajectory (blue) with fixed-point candidates (red).

(b) All fixed-point candidates (red) overlaid on trajectory paths (gray).

(c) Multiple trajectories projected onto the same 3D space.

Figure C.1 Hidden state dynamics of the trained IntentionRNN on the labyrinth dataset, projected onto the first three principal components (95% variance explained). Fixed-point candidates are timesteps where the hidden state step norm $\|h_t - h_{t-1}\|$ falls below the 5th percentile.

tions for future work.